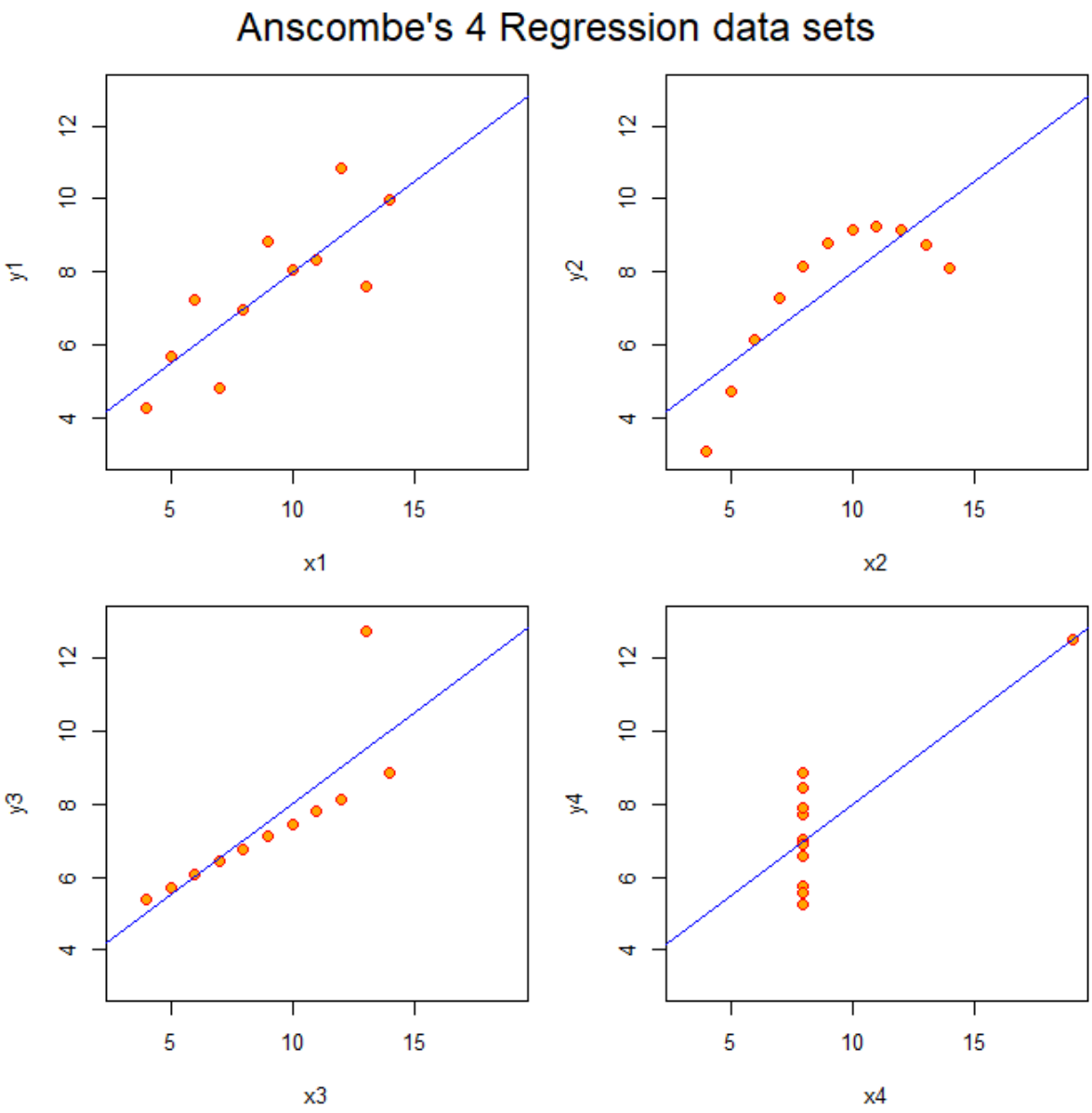
Always Plot Before You Regress!

When constructing and estimating any statistical model it is highly advisable to "appraise" the data that are to be used before proceeding. With extremely large data-sets, this in itself can be challenging. However, in many cases this appraisal can be based on some relevant plots. Failure to do this can have serious consequences as a result of inappropriate modelling.

Anscombe [1] made this point in a very clear manner by considering simple regression models for four data-sets, since referred to as "Anscombe's Quartet". Here are the fitted least squares results:



For each of the different data sets the fitted regression line (with standard errors in parentheses) is

$$\hat{y}_j = 3.00 + 0.50x_j \quad ; \quad j = 1, 2, 3, 4 \quad ; \quad R^2 = 0.67$$
$$(0.12) \quad (0.12)$$

The data are available in the function 'anscombe' in the 'datasets' package [2] in the base R software [3]. The R code used to produce the above plots can be downloaded from [4]. Adding the command summary (mods[[1]]) will produce a standard regression summary for the first model, and similarly for models 2, 3, and 4.

These plots illustrate (a) that a fitted linear regression line may be consistent with several different linear or non-linear data sets; and (b) outliers in the sample can dramatically alter a fitted linear regression. *The key message here is that it is vitally important to plot the data before fitting any regression model*.

Here is a similar example that I used to use when teaching introductory regression analysis, to convey the same message. Let's begin with the following R code:

```
x<- c (1.5,2,2.5,2,2,2,1.7,1.5,1.3,3.5,3.5,3.5,3.7,3.9,4.1,4.3,4.5,4.5,4.5,5.5,5.5,5.5,5.5,
  6.5,6.5,6.5,6.5,6.0,5.75,6.25,7.5,7.5,7.5,7.5,7.75,8.0,8.25,8.0,8.25,8.5)
y<- c (5,5,5,4,3,2,1.7,1.8,2.0,8.0,7.0,6.0,5.5,5.3,5.3,5.5,6.0,7.0,8.0,12.0,11.0,10.0,9.0,
  12.0,11.0,10.0,9.0,10.5,11.5,9.5,15.5,14.5,13.5,12.5,14.0,14.5,15.0,13.5,13.0,12.5)
model<- lm (y ~ x)
summary (model)
```

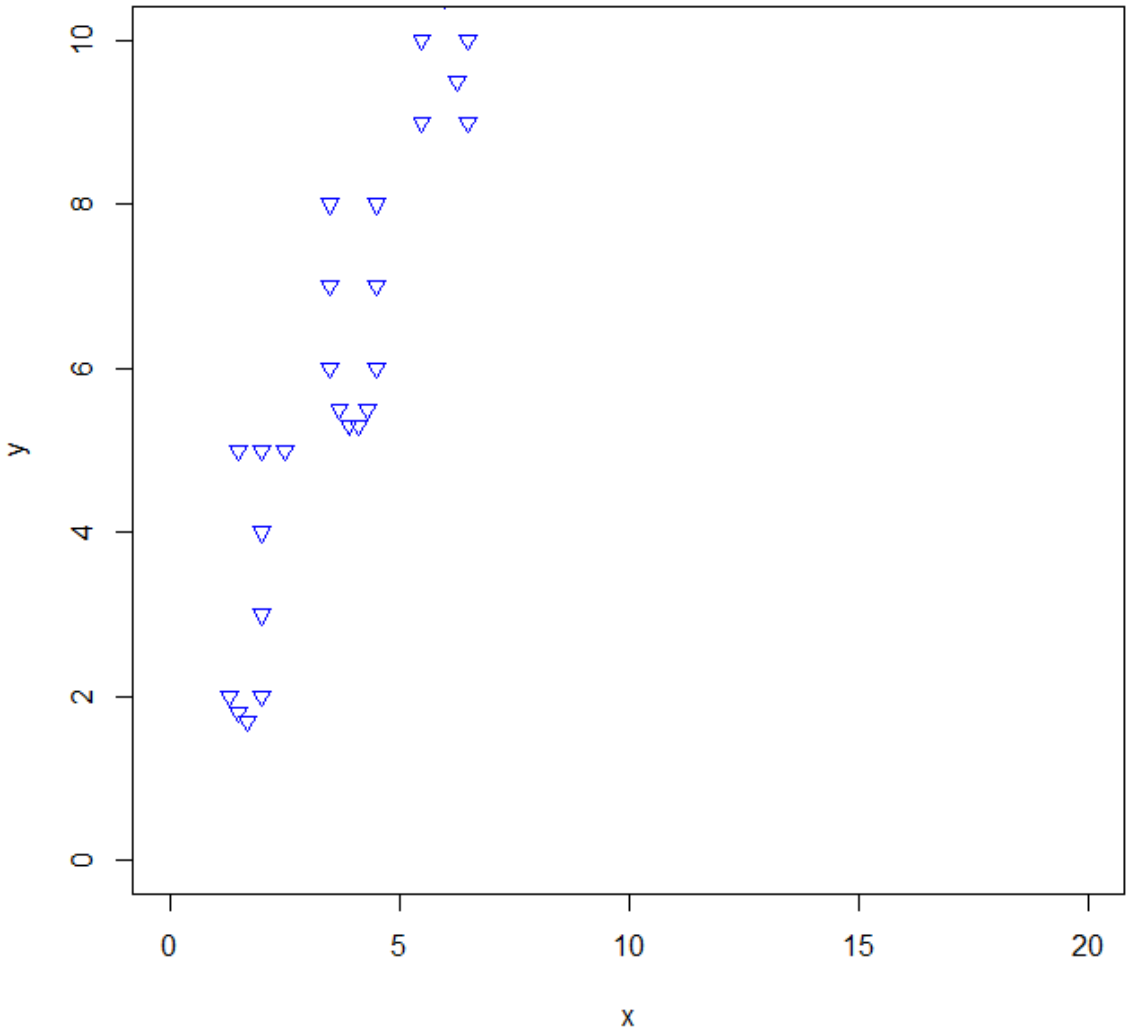
The associated output is:

```
Residuals:
    Min      1Q  Median      3Q     Max
-2.2214 -1.2112 -0.1202  0.7554  2.5466

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.03664    0.52540   -0.07    0.945
x            1.73200    0.09542   18.15   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.35 on 38 degrees of freedom
Multiple R-squared:  0.8966,    Adjusted R-squared:  0.8939
F-statistic: 329.4 on 1 and 38 DF,  p-value: < 2.2e-16
```

The standard error and t-statistic associated with the regressor indicate exceptional "significance", and the R^2 of 0.8966 suggests a excellent "goodness of fit".

Oh! We forgot to plot the data prior to fitting the regression model. Let's rectify that:

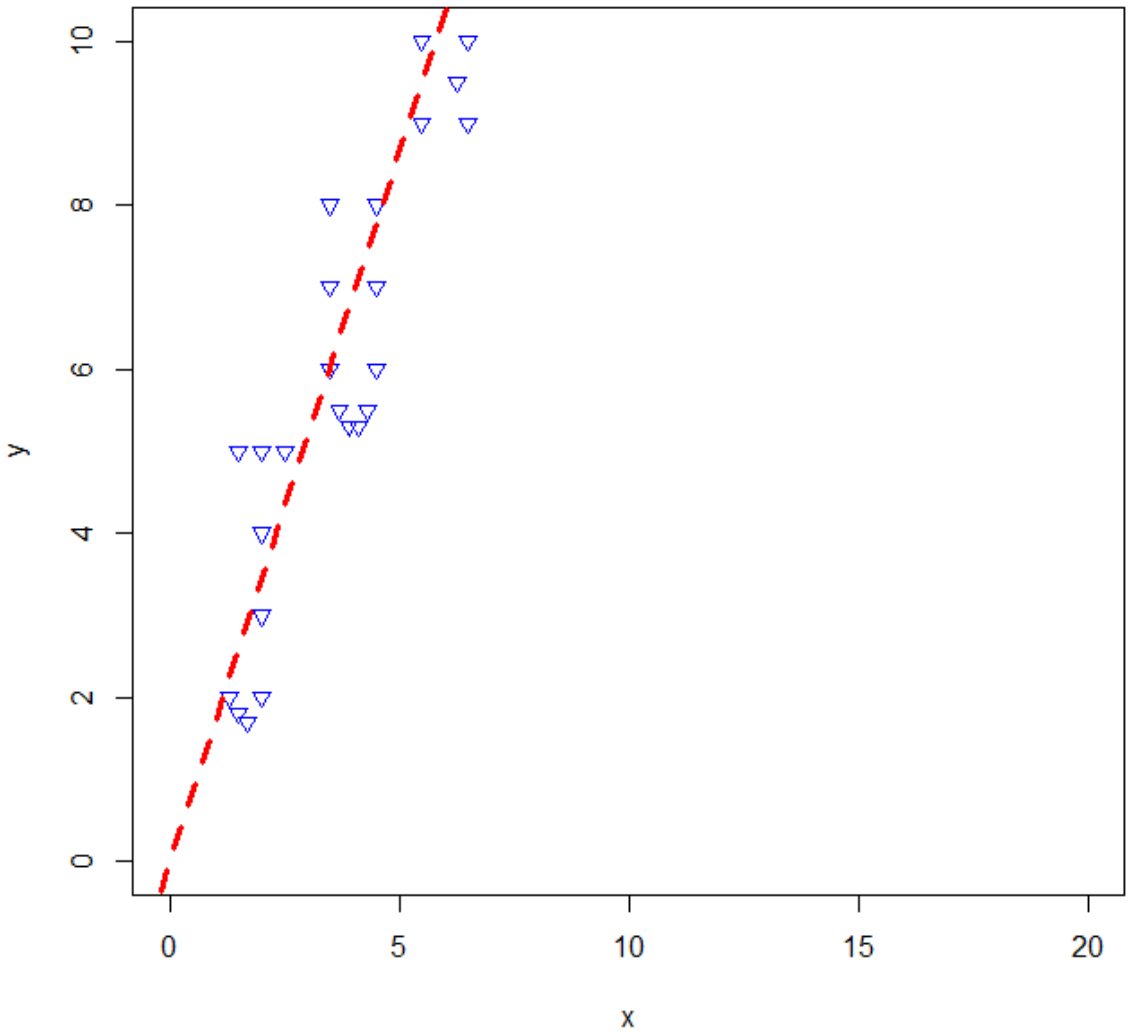
```
plot (x, y, pch=25, col="blue", xlim=c (0,20), ylim=c (0,10))
```



We can add the fitted OLS regression line with command:

```
abline (model, col = "red", lwd = 3, lty = 2)
```

resulting in the following plot:



At a first glance, this looks very reasonable – it's similar to the first of Anscombe's results in that the data points appear to be quite randomly scattered about the *linear* fitted line, with no apparent outliers.

So, maybe it didn't matter that we estimated the regression model before plotting the data? Well, simply plotting the data is one thing, but plotting it in a meaningful manner is another. Here, the last plot is inappropriate – the scales of the 'x' and 'y' axes are chosen very poorly. Looking back at the data on p.2 we see that the sample values for 'x' range from 1.3 to 8.5 (*not 0 to 20*), while those for 'y' range from 1.7 to 15.0 (*not 0 to 10*)!

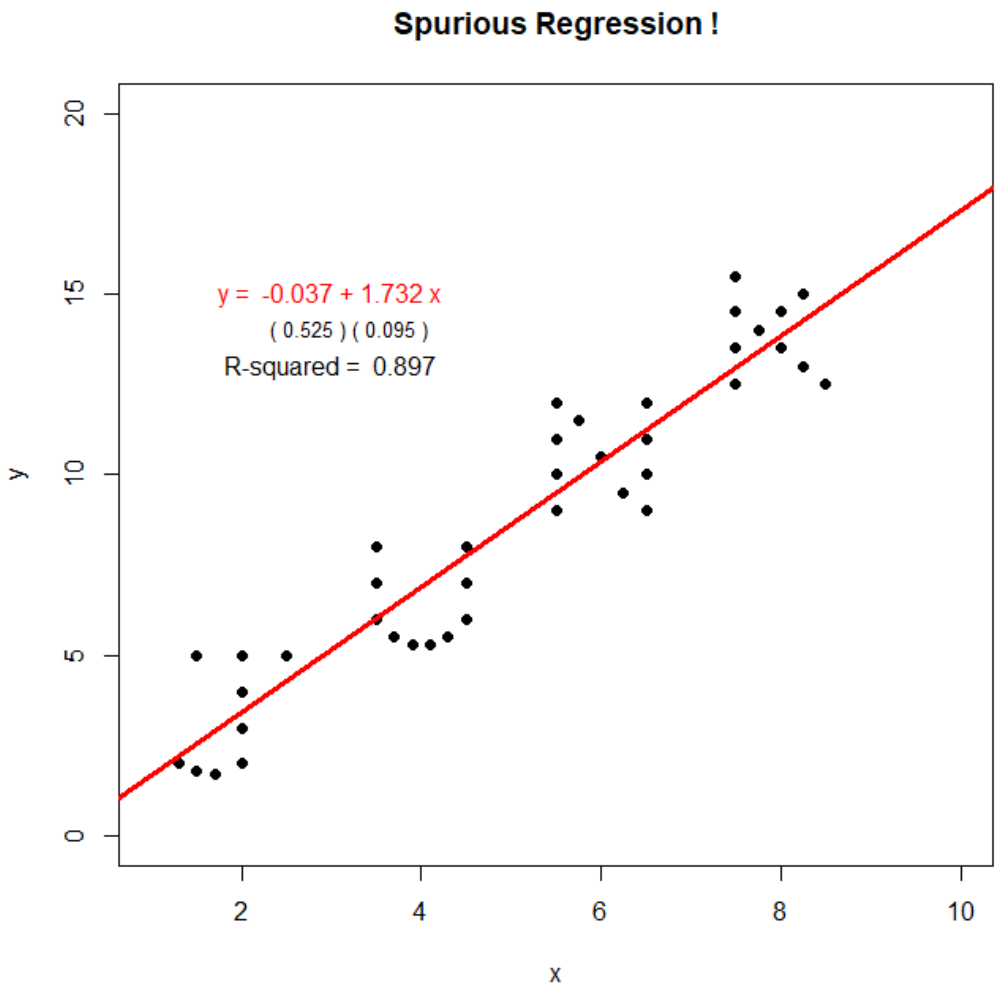
Let's fix that problem by using an x-axis scale of 0 to 10, and a y-axis scale of 0 to 20, so that the sample is fully covered:

```
plot (x, y, pch = 16, xlim=c (1,10), ylim=c (0,20), main = "Spurious Regression!")
abline (model, col = "red", lwd = 3)
```

Also, let's add a few more details to the plot:

```
coef <- round(coef(model), 3)
se<- round(sqrt(diag(vcov(model))),3)
text (3, 15, paste ("y = ", coef[1], "+", coef[2], "x"),col="red")
text (3.2,14, paste ("(", se [1],")","(", se [2],")"), cex=0.8)
rsq<- round(summary(model)$r.squared, 3)
text (3,13, paste ("R-squared = ", rsq))
```

Here is the result:



Well, the t-statistic may have been fabulous, and the R^2 value may have been very appealing, but if we had plotted the data (*properly*), then we probably wouldn't have even considered fitting a least squares regression at all!

You can download the data and R code for this "spurious regression" from [5].

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References

[1] Anscombe F. J., 1973. Graphs in Statistical Analysis. *The American Statistician*, 27(1), 17–21.

[2] https://www.rdocumentation.org/packages/datasets/versions/3.6.2

[3] R Core Team, 2021. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org/

[4] https://stat.ethz.ch/R-manual/R-devel/library/datasets/html/anscombe.html

[5] https://github.com/DaveGiles1949/r-code/blob/master/Plotdata.R